

MACHINE LEARNING APPLICATIONS IN FINANCIAL LEASING

Group Id - 25-26J-179

Project Proposal Report

Maharambage K.N– IT22129130

B.Sc. (Hons) Degree in Information Technology
Specialization in
Information Technology

Department Of Information Technology

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Supervisor - Mrs. suryaa kumari

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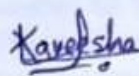
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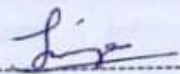
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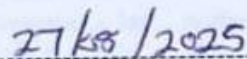
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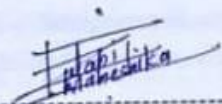
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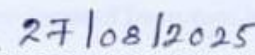
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Abstract

Accurately predicting customer default risk poses significant challenges in the financial leasing industry which directly affects financial profitability, income stability, and sustainability of a long term business. Current traditional credit scoring systems, which primarily draw on static financial data fail to account for behavioral patterns that change and signal movement before customers default on a payoff agreement. It's clear that any lag in risk identification or excessive increase in non-performing loans (NPLs) due to unexpected defaults would reinforce assumptions about the current models' scoring validity which would lead to further deteriorating financial results. The current research project proposes to develop a machine learning model that is accurate and interpretable, to forecast default risk in vehicle leasing clients, while also incorporating traditional financial metrics alongside behavioral metrics into the prediction model. This will include information from leasing company databases on customers including not only financial history and payment history, but also their employment history and transaction behavior. A domain-aware data preprocessing pipeline will examine potential issues that are known to accompany transaction data typical such as omitted values, differences in label terminology and dates, and lagged data entry; this is prior to deciding upon which machine learning classifiers to implement in the model. In the feature engineering phase of the project behavioral-enriched variables associated with historical financial behavior meant to forecast future outcomes by evaluating payment, transaction, and customer historical latency factors will be included. A range of supervised classification algorithms (e.g., decision trees, random forests) will be trained and evaluated to allow the best option based on accuracy and explainability before implementing into a web based decision support system to provide real-time forecasts.

The system proposed will use advanced analytics combined with explainable AI methods, allowing leasing companies to quickly identify the most high-risk clients, allow companies to make compliant, reasoned decisions, and support proactive credit risk management. The objective of the proposed system is a decision tool that enables increased productivity, lower costs associated with default loss, and improved business viability through future-focused, data driven risk evaluation.

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List Of Abbreviations

Table 1-List of abbreviations

Abbreviation	Description
NPLs	Non-Performing Loans
ML	Machine Learning
XAI	Explainable Artificial Intelligence
PD	Probability of Default
AI	Artificial Intelligence
MM	Mass Market clients
PM	Premium Market clients

1. Introduction

One of the main issues regarding financing leasing is managing customer default risk because it is likely to affect profitability, capital position, and the firm's long-term viability. In Sri Lanka, vehicle leasing companies have historically applied static credit-scoring models that, by and large, focus on assessing applicants with respect to their previous financial history and demographic information. Thus, although the credit-scoring model served as an overall descriptive risk model, it did not consider behavioral risk, such as late payments, non-payments and inconsistent payments, that may indicate the customer may default [1]. The limitation of the model may result in a delayed response to dealing with the risks and the growing risk of non-performing loans (NPL), which generally results in an increase in operational risk and financial risk [2].

In recent years, machine learning (ML) developments have shown huge promise to improve borrowing prediction by utilizing augmented data sources and capturing complex non-linear interactions [3]. Many classification models have been utilized to predict default risk, including logistic regression, random forests, gradient boosting, and neural networks - showing improved performance compared to traditional scorecard approaches [4]. In addition, with the emergence of Explainable AI (XAI) methods, they address the interpretability issue allowing model results to be more interpretable, transparent, and justifiable - a necessity when working in regulated financial contexts [5]. However, despite this global progress, ML-based behavioral credit risk models have not been widely implemented in Sri Lanka's vehicle-leasing sector, and institutions remain limited to static financial scoring systems [6].

This study attempts to resolve that gap by building an accurate and interpretable ML classification model for default risk prediction for vehicle leasing purposes. In this case, we will provide a user a system that will enhance the traditional financial features with borrower behavioral features – or characteristics such as debt-to-income ratios, variance and stability of employment history, trends of overdue payments, and irregular transaction patterns – in order to achieve a more robust estimate of the probability of default. By doing specific feature engineering in the domain and creating tailored preprocessing pipelines, we will also make use of those noisy and inconsistent financial data. The last prediction model will serve as a component of a web-based decision-support system enabling real-time risk evaluation of loans and proactive portfolio planning with the goal of having a more resilient vehicle leasing capability in Sri Lanka using early warning assessment mechanisms combining ML with behavioral risk enrichment.

1.1 Background & Literature survey

Management of credit risk is vital for financial stability and profitability in the vehicle leasing space. Vehicle leasing companies have relied on a static, rule-based process for credit scoring and rely mainly on historical financial information. While rule-based approaches generate relatively consistent risk assignments from a credit risk score, they may miss the instinctive behavioral cues—such late payments, part settlements, or out of character transactions—that could signal increased default in the future, well before a customer is ultimately considered non-performing [7], [2]. The absence of a pre-default detection mechanism - through machine learning systems - contributes to non-performing loans (NPL) that inhibits ability, or institutional resiliency to retain and maintain earnings, limit risk, reduce or avoid procedural inefficiencies, and also challenge financial regulators [1], [8].

Machine learning (ML) modeling is well suited to model credit risk assessment in all forms by latching onto the evolving, non-linear impacts of, and relationships between, large and disparate datasets. Classification based algorithms, in particular, excel at default prediction; they can, in fact, yield better estimates than traditional scorecards with probability-of-default (PD) estimates [4], [9]. By relying on a combination of financial factors- income, total liabilities, debts-to-income resource ratios, further derived from collateral values-values are all behavioral cues on creditworthiness for MM and PM clients [6], [10]. A vehicle leasing library of behaviors-such trending inequities in payment over the last fifteen years-can provide further signals of client financial resiliency beyond predicting default, as well as clues related to signaling contextual awareness of market risk undermining client ability [10].

There are also operational influences affecting default risk outcomes. Consideration for the management practices, how information is passed through the organization, and how they are incentivized can be important factors along with data quality within leasing car operations related to how, when and if default risk matters are detected and reported [1] [8]. It should be evident that harnessing predictive models against organization or context or management practices are as important as the data related to the borrower [3], [6].

The proposed Default Risk Prediction model uses classification methods to predict the likelihood of a vehicle leasing customer becoming non-performing. It introduces behavior-based features, financial features, and changes in behaviors over time purposefully, in order to provide explainable outputs (using XAI approaches) [4], [11]. This risk probability information can be used in a wider view of decision-making before committing, to inform provisioning models, determining branch performance measures and reporting, and ultimately informing the customer retrospective segmentation strategy when customers are categorized as Low, Ideal, Reasonable, and/or High Default Risk for credit risk management that is proactive and forward-looking [6], [4], [10] .

1.2 Research Gap

When reviewing the academic literature related to credit risk modeling in the financial leasing space, it is evident that most of the research relies on historical financial statements and uses traditional credit scoring approaches with limited emphasis on changing behavioral actions. There have been a number of models which have included machine learning as a method for estimating credit risk [12], [13], [14] but many of the past research did not include behavioral-based metrics, including late to pay or only repaying partial amounts, and inconsistently scanning records. This is a substantial gap in the literature to ignore as these behavioral measures could serve as precursor indicators of borrower default that may be useful for taking preventative action to avoid defaults in the first place.

Another gap of the research has to do with the borrower behavior being time dependent. Time-series modeling has been used in a number of contexts, however, I have not seen default prediction models use dynamic feature engineering to track the directional aspects of borrowers behavior over time in the vehicle leasing industry. Without this time-dependent context, models may not pick up any deteriorating behaviors early enough to take preventative action. With respect to data preparation, much of the past research does not discuss how to overcome the issues of last-on, first-off specific inconsistencies in the leasing space such as the delay in withdrawals, repeated missing values, and the inconsistent capture of data from branch-level systems. The lack of tailored data cleaning pipelines to deal with original conditions can also impact model output accuracy and reliability [15].

We should add another area that has been lacking coverage. Most credit risk models are designed for periodic reporting and processed in batch mode, rather than real-time, web-based decision support. This effectively means that there is a gap on the operational applicability of models, as leasing companies need instant risk decisions and assessments at loan origination and while monitoring the changing conditions of clients in a generated (or live) portfolio.

The solution being proposed in this investigation will close these gaps by,

- Integrating financial and behavior features for enhanced early default detection.
- Dynamic feature engineering to incorporate repayment trends over time.
- Specialized data cleaning workflows for data anomalies unique to the leasing industry.
- The model can be employed as a real-time web-based decision support system.
- Outputs can be explainable and also meet regulatory compliance, thus complying with transparency.

By closing these gaps, the system will empower leasing companies to detect at-risk clients earlier, reduce the risk in financial provisioning, and ensure portfolios remain being stable in a rapidly changing financial environment [12].

Table 2-Research gap

Feature / Criteria	Reference [12]	Reference [13]	Reference [15]	Reference [14]	Proposed Solution
Inclusion of behavioral-enriched features (overdue payments, partial settlements, irregular transactions)	X	✓	X	✓	✓
Dynamic feature engineering (temporal trends in repayment behavior)	X	X	X	X	✓
Customized data cleaning for leasing-specific irregularities	X	X	X	X	✓
Customized data cleaning for leasing-specific irregularities (missing values, delayed updates)	X	X	X	X	✓
Real-time risk prediction (web-based decision support)	X	X	X	X	✓

1.3 Research problem

The vehicle leasing industry plays an important role in facilitating personal mobility and doing business, and is a contributor to expansion in financial services. However, an ongoing risk to profitability, capital adequacy, and a sustainable future, is its inability to measure and proactively predict those customers that may ultimately default on their lease payments or finance arrangements. Many leasing companies have used static, rules-based credit scoring systems that use historical financial information (such as profitability, financial strength, and existing credit) to measure risk and the issuers of credit have relied on these traditional measures for too long [1]-[4]. While traditional financial metrics do provide a foundation for risk decisions, behavioral metrics such as partial payments, late payments, and transaction irregularity describe a customers ongoing ability to repay their debts before becoming a non-performing account [2], [10], [14].

The inability to measure dynamic, data driven customer risks leaves the financial services sector exposed to an increase in non-performing loans leading to write-offs, increased provisioning and decreased capital adequacy. The dynamic and contextual nature of the risks associated with lending means that a lack of data driven clearly articulated risk measures limit the ability to take preventative action (such as applying different lending terms) once risk information has been assimilated, and limit the ability to provide additional scrutiny in assessing the risk at that time, and to be able to offer a solution (for example a formal repayment arrangement) to modify the risk and future repayment potential with financially vulnerable clients [2], [3], [10], [15]. The lack of reasoning which is required from many predictive and interpretative models may deter the proposed engagement and use for decision-making and compliance .

To remedy this significant gap, this research proposes to develop an accurate and interpretable machine learning (ML) model that predicts vehicle leasing client default risk factors by combining financial and behavioral data [6], [11]. The solution will also have dynamic feature engineering to reflect repayment behavior trends over time, domain-specific data cleaning pipelines to resolve inconsistencies, and Use Explainable AI (XAI) to develop transparency and trust in predicted outcomes [5], [11], [15]. Once model outputs are implemented in real-time, web-based decision support systems, lease organizations will get stronger performance management by detecting a high-risk client sooner, managing resource allocation even sooner, and employ credit risk management in forecasting [6], [12]- [14].

Finally, this research expects to drive down unanticipated defaults, limit commercial losses, and increase the long-term viability of vehicle leasing organizations and other entities. Addressing gaps between traditional static scoring processes and dynamic, analytic behavioral analytics will allow leasing organizations to change how they engage default risks while becoming operationally proactive in increasingly competitive capital markets [1], [7], [10], [14].

2. Objectives

2.1 Main objective

The said idea of this research is to shape the credit risk management of the vehicle leasing industry through advanced machine learning. The research proposes to develop a more precise and interpretable predictive model that takes into account both financial and behavioral information in predicting customer default risk. The research ultimately seeks to develop a model that helps leasing companies proactively, as opposed to relying on static, rules-based, scoring methods. It will also anticipate dynamic credit risk to provide a decision making framework in real-time to improve financial resilience and deter non-performing loans (NPLs). The model will use classification algorithms, dynamic feature engineering, and XAI (explainable artificial intelligence) techniques to ensure risk predictions are accurate, interpretable, and auditable. The goal will be to help leasing companies identify high risk clients in advance, operationalize risk to optimize human resources, and develop real-time, proactive approaches to lending, service delivery, operational efficiency, and business sustainability.

2.2 Specific objectives

- To gather, clean, and preprocess financial and behavioral data of vehicle leasing customers for use in machine learning models.
- To develop temporal features and domain-specific features that capture the functional change in repayment behavior.
- To create and evaluate ML classification models that accurately forecast default risk and are interpretable.
- To apply Explainable AI (XAI) methods for transparency, auditability, and regulatory compliance.
- To deploy the ML model as a decision support, web-based system that supports real-time, forward-looking credit risk management.

3. Methodology

3.1 System Diagram

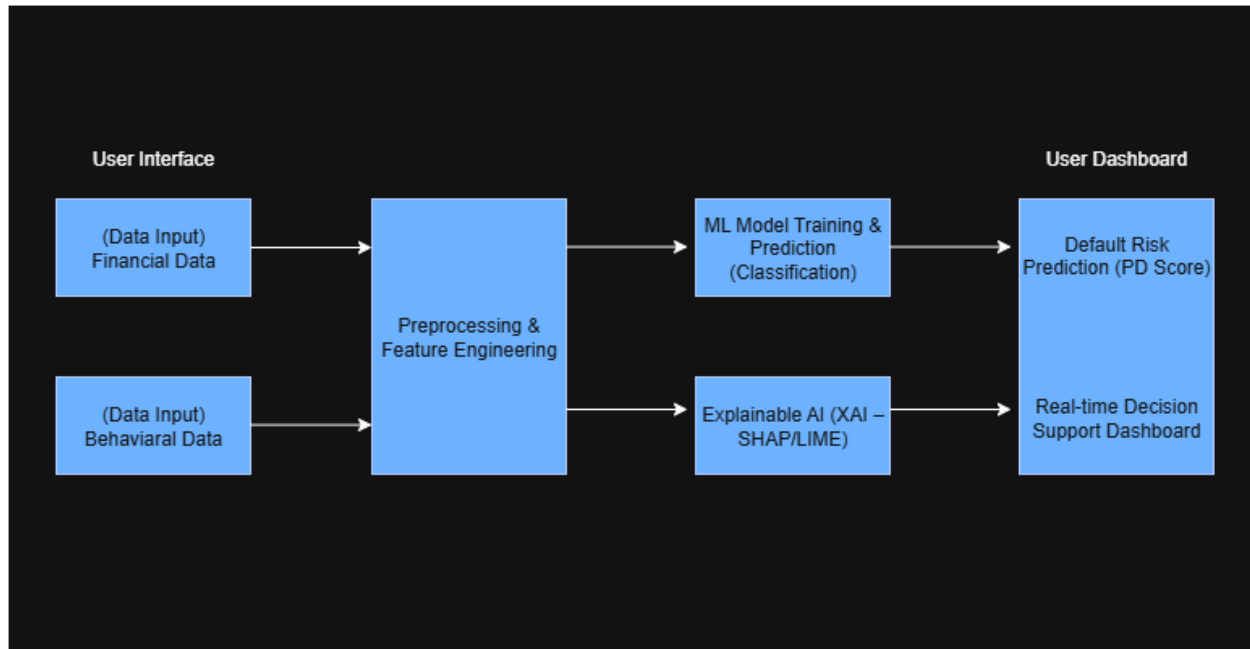


Figure 1-System diagram

The system diagram displays a high-level overview of the structure of the Default Risk Prediction feature of our application, which is devised to assist vehicle leasing companies to capture potential defaults prior to defaulting. The system has the capacity to use financial and behavioral data, applies best in class machine learning (ML) capabilities, and produces results on a real-time decision-support dashboard. We will describe each component and how they interact with what follows.

- (Data Input) Financial Data:

This is the starting point of the system where structured customer financial records (income, debt-to-income ratio, history of credit, and other loans) are provided as inputs. The financial indicators represent conventional predictors of credit risk.

- (Data Input) Behavioral Data:

In addition to financial records, behavioral information (late or partial payments, transactions after normal spending patterns, changes in spending behavior, etc.) are captured as well. The responsiveness of behavioral indicators provide an additional layer of intelligence to the prediction of default, as these indicators typically indicate early warning signs of possible default.

- Processing & Feature Engineering

This module is responsible with cleaning, transforming and enriching the raw data. A check is made for missing values, inconsistencies or delays. We also create features such as lags for missed payments, employment stability, payment trends, etc. to capture time-based risk. We combine both financial & behavioral features into a unified dataset that machine learning models can act on.

- ML Models training & predictions (Classification)

The processed dataset is fed into machine learning algorithms (example: Logistic Regression, Random Forest, XGBoost, etc.) and customers are classified into risk categories, (low, medium or high probability of default).

The model not only predicts Probability of Default (PD Score) but also learns and improves with new data over time. This allows leasing companies to report and monitor risk understanding in real-time before approving new leases. Explainable AI I include Explainable AI tools such as SHAP (Shapley Additive explanations) and LIME (Local Interpretable Model-agnostic Explanations) to ensure transparency and trust in the prediction. Explanations identify which factors (for example, overdue payments or high debt ratios) can influence a model's decision. I always strive to make the model interpretable so that decision-makers can incorporate it in the decision-making process and comply with regulatory procedures in financial services. User dashboard

- Default Risk Prediction(PD Score)

The central output is the Probability of Default (PD Score) of each customer. The PD Score allows financial managers to quantify the predictability of a customer's transition into a non-performing loan (NPL), enabling managers to act and respond proactively.

- Real-time Decision Support Dashboard

I create a comprehensive dashboard to provide a user interface for risk managers and decision-makers tool. The dashboard will provide customers risk profiles, PD scores, and early warning indicators. Scenario and what-if simulations will be provided, allowing a user to test circumstances with different financial statements. Portfolio monitoring will also available so customers will dynamically adapt their strategies and resources over time.

3.2 Gantt Chart

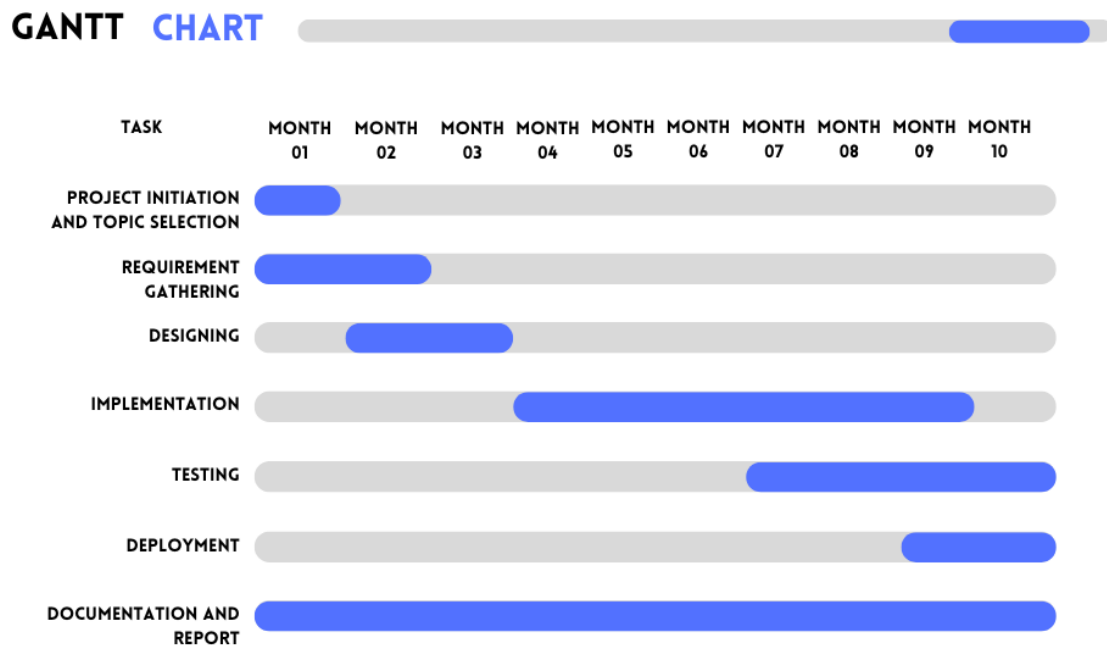


Figure 2-Gantt chart

3.3 Work breakdown structure

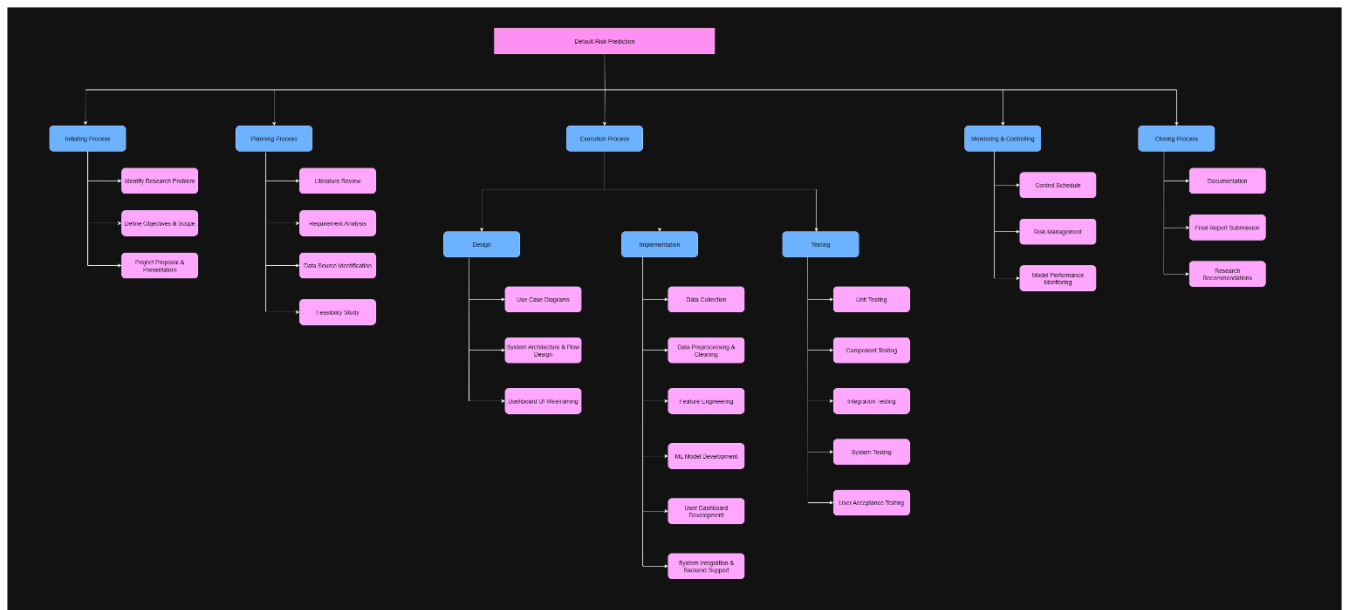


Figure 3-Work breakdown structure

4. Project Requirements

4.1 Functional Requirements

- Data Input Capability

Enable the system to safely and securely access and extract data from customer financial data and behavioral data (i.e., income credentials, credit history, debt to income, and lags in payment, partial payment, irregular transactions output source data) from leasing company data repositories.

- Preprocessing & Feature Engineering

Design automated methods for the data cleansing of raw financial and behavioral data; extract sample sources and solutions for how to resolve missing data; and recommend features based and generated by the leasing domain such as Lagging payments, job length, and patterns of payment occurrence.

-
- Machine Learning Model Training & Prediction

Deploy classifier-based ML models to calculate the probability of default (PD - ML output) for each leasing client. Be certain that the classification model has good prediction accuracy, can be scaled, can classify semi-structured and structured data (sometimes a data source may not be in the structured form) and, is reputable (h6).

- Real-Time Decision Support Dashboard

Provide a web-based interactive dashboard where risk officers can view default risk scores (i.e., predictable behaviors), compare default risks using various time horizons, view and track the portfolio level risk exposure management, drill-down filtering of clients by alternative risk classifications used for decision making.

- Integration with Risk Management Systems

Make sure, the output of the predictive model (i.e., prediction of credit loss) can easily integrate with existing systems of a leasing company (e.g., Expected Credit Loss (ECL) estimating tools & branch performance management systems/products) for the very purpose of assessing your forward-looking credit risk capability.

4.2 Nonfunctional requirements

- Performance

The system must be able to process customer financial and behavioral data and generate a predictive model of default risk, within a bought time frame of 3-5 seconds per request. The performance of the model will need to ensure timely decisions are made in the lease approval process.

- Scalability

The system must be able to handle large amounts of customer data (10s-thousands of records) and allow for multiple users (e.g. credit officers, branch managers) to access the system concurrently without a degradation in performance.

- Security

Data confidentiality and data integrity must be enforced via encryption (AES-256 for storage, TLS for transmission), role-based access controls, additional logging capabilities and compliance with financial regulatory requirements (examples: Data protection practices in Sri Lanka must be compliant with GDPR).

-
- Reliability & Availability

The application must have no less than 99% uptime. The application must be designed with fault tolerance and be backed-up regularly to mitigate data loss in the event of application/system failure.

- Maintainability

The system must be coded using a specific set of modular standards, including sufficient documentation, to allow future developers to extend or troubleshoot the ML models and dashboard easily.

- Usability

The dashboard must be designed with usability; for financial officers who do not have advanced technical expertise. Graphical data visualization including graphs, PD scores, and time trends must be comprehensible and must appear interactive.

4.3 User Requirements

- Accessible

Credit and loan officers, branch managers, and financial analysts should be able to access the system easily on a secure web-based dashboard without significant tech knowledge.

- Reliable Predictions

The system should provide reliable and accurate default risk predictions (Probability of Default - PD score) for vehicle leasing clients, so that financial institutions may assess credit risk when considering client credit applications.

- Real-Time

Users should receive predictions and risk explanations in real-time (real time definition: seconds after submitting information) to allow for quick decision-making relating to processing applications for loan requests.

- Explainable Predictions

The system must provide explainable predictions (SHAP/LIME visualization) for each prediction, so that users can understand why a client is classified as high or low risk.

4.4 System requirements

Tools and Technologies

Tools & Technologies	Purpose
React	For frontend development
HTML5, CSS3, TypeScript	For structuring, styling, and scripting frontend
.NET Core	For backend development and APIs
REST API	For communication between frontend and backend
Python (scikit-learn, TensorFlow, PyTorch)	For machine learning and data science tasks
Pandas, NumPy	Data handling and analysis
Matplotlib, Seaborn	For data visualization
SQL Sever / MY SQL	For database management
Jira	For project management
Git & GitHub	For version control
Microsoft Teams	For team communication
Compliance & Standards	For following financial standards
MS Word	For reporting and documentation

Table 3-Tools and Technologies

5. Budget and budget justification

Category	Estimated Cost	Justification
Computing Resources (Lab PCs)	free	Using university lab computers/servers
Software & Tools	free	Using free/open-source tools
Data Preprocessing	free	Research assistant/student effort to clean 180,000+ rows and 10+ columns.
Model Development & Testing	Rs.5000	Cost for building, tuning, and testing ML models for prediction
Statistical & Financial Analysis	free	Tasks related to data analysis, focusing on IFRS 9 compliance, as well as feature selection.
Dissemination (Reports / Printing)	Rs.1500	Printing, report preparation and publication costs
Contingency (10%)	Rs.500	Extra budge

Table 4-Budget and budget justification

6. References

- [1] A. G. R. R. P. T. Thomas, "Credit scoring models and risk management in financial services: A review," *Journal of Risk and Financial Management*, vol. 14, no. 7, p. 1–18, Jul 2021.
 - [2] H. P. C. K. Wijesundara, "Non-performing loans and risk factors in Sri Lankan leasing industry," *Asian Journal of Finance & Accounting*, vol. 12, no. 2, p. 87–103, 2020.
 - [3] B. B. H.-V. S. L. C. T. E. Lessmann, "Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research," *European Journal of Operational Research*, vol. 247, no. 1, p. 124–136, 2015.
 - [4] N. G. P. J. H. W. J. B. Heaton, "Deep learning for finance: deep portfolios," *Applied Stochastic Models in Business and Industry*, vol. 33, no. 1, p. 3–12, 2017.
 - [5] I. Ž. T. C. F. Kamiran, "Quantifying Explainability and Trust in Machine Learning Systems," *IEEE Transactions on Knowledge and Data Engineering*, vol. 33, no. 10, p. 3442–3454, 2021.
 - [6] D. P. N. Jayasekara, "Machine Learning Adoption in Credit Risk Analysis: Evidence from Sri Lankan Financial Institutions," *Sri Lanka Journal of Economic Research*, vol. 9, no. 2, p. 15–32, 2021.
 - [7] D. P. N. Jayasekara, "Machine Learning Adoption in Credit Risk Analysis: Evidence from Sri Lankan Financial Institutions," *Sri Lanka Journal of Economic Research*, vol. 9, no. 2, p. 15–32, 2021.
 - [8] H. P. C. K. Wijesundara, "Non-performing loans and risk factors in Sri Lankan leasing industry," vol. 12, no. 2, p. 87–103, 2020.
 - [9] B. B. H. V. S. L. C. T. E. Lessmann, "Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research," *European Journal of Operational Research*, vol. 247, no. 1, p. 124–136, 2015.
 - [10] B. B. H. V. S. S. Lessmann, "Behavioral scoring and credit risk: Evidence from machine learning approaches," *Expert Systems with Applications*, vol. 105, p. 1–15, 2018.
 - [11] S. B. G. J. M. d. S. L. R. Vargas, "Explainable AI for credit risk: Combining SHAP values with traditional credit scoring," *Journal of Financial Data Science*, vol. 4, no. 3, p. 45–59, 2022.
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- [12] S. K. D. T. R. Nyman, "News and narratives in financial systems: Exploiting big data for systemic risk assessment," *Journal of Economic Dynamics & Control*, vol. 127, p. 104119, 2021.
- [13] Y. M. M. U. N. P. Appiahene, "Predicting bank operational efficiency using machine learning algorithm: Comparative study of decision tree, random forest, and neural networks," *Advances in Fuzzy Systems*, vol. 2020, 2020.
- [14] L. Pierce, "Organizational structure and the limits of knowledge sharing: Incentive conflict and agency in car leasing," *Management Science*, vol. 68, no. 4, p. 1822–1842, 2022.
- [15] M. A. Faheem, "AI-driven risk assessment models: Revolutionizing credit scoring and default prediction," *IRE Journals*, vol. 5, no. 6, p. 53–58, 2021.

7. Appendices

- System diagram
https://drive.google.com/file/d/1k9yZxZ_wIti_e3PgS5aW5H8rUW8yms1d/view?usp=drive_link
- Work breakdown structure
https://drive.google.com/file/d/1QwU8EYFKy477Z56peW9NB6MAc2eBaJER/view?usp=drive_link



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